

Module 01: Surfaces and Spaces: The Topology of Crocheted Fabric

Learning Objectives

1. Identify how crochet creates topological surfaces with distinct curvature properties — flat, hyperbolic, and spherical — through stitch increase and decrease patterns.
2. Recognize that every piece of crocheted fabric is a network (graph) where stitches are nodes and connections are edges, and relate this graph structure to neural network architecture.
3. Map the topological concepts of Euler characteristic, Gaussian curvature, and boundary to the Active Inference concepts of Markov blankets, generative models, and system structure.

Introduction

In 1997, mathematician Daina Taimina picked up a crochet hook and did something that had puzzled geometers for over a century: she built a physical model of hyperbolic space.

Mathematicians had proven that surfaces of constant negative curvature existed, but paper models tore, and computer renderings stayed trapped behind screens. Crochet succeeded where other methods failed because the craft builds surfaces stitch by stitch, row by row, with no requirement that the fabric stay flat. A crocheter who adds too many stitches per round does not make a mistake — she makes a hyperbolic plane.

This insight is the doorway into our module. Every piece of crochet fabric is a **surface** with **topological properties**. The flat dishcloth, the ruffled scarf edge, the rounded amigurumi bear head — these are not just pretty objects. They are embodiments of mathematical surfaces with measurable curvature, classifiable topology, and network structure that maps directly onto the architecture of neural networks.

In this module, we will look at crocheted fabric through the lens of topology (the mathematics of surfaces and spaces), graph theory (the mathematics of networks), and Active Inference (the

framework that connects both to how agents perceive and act in the world). You do not need to know any of these fields in advance. You already know them — your hands have been doing topology every time you picked up a hook.

Key Concepts

1. Crochet as Topology Made Tangible

Topology is the branch of mathematics that studies properties of shapes that do not change when the shape is stretched, bent, or deformed — only when it is cut or glued. A coffee mug and a donut are topologically the same (both have one hole). A sphere and a cube are topologically the same (neither has a hole). Topology cares about connectivity, not measurement.

Crochet is one of the few crafts that naturally creates surfaces of different **Gaussian curvature** — the intrinsic measure of how a surface curves at each point.

Zero curvature (flat): When you crochet a flat circle using the standard formula — start with 6 single crochet in a magic ring, then increase by 6 evenly spaced stitches each round — the fabric lies flat on the table. It has zero Gaussian curvature, like a sheet of paper. The increases exactly compensate for the expanding circumference. Each round has 6 more stitches than the last: 6, 12, 18, 24, 30. The geometry works out perfectly. Lay it on the table and it sits there, calm and well-behaved.

Negative curvature (hyperbolic): Now imagine you increase by 12 stitches per round instead of 6. Or increase in every stitch instead of every other stitch. The fabric has too many stitches for its circumference. It cannot lie flat. It ruffles, waves, and folds over on itself — a lettuce-leaf edge, an ocean coral frill. This is **negative Gaussian curvature**, the hallmark of hyperbolic geometry. The surface is saddle-shaped at every point. Taimina's breakthrough was realizing that this "mistake" — too many increases — was actually a precise physical model of hyperbolic space.

Positive curvature (spherical): What happens when you increase too slowly — say, only 3 stitches per round instead of 6? The fabric runs out of stitches for its radius. It cups inward, forming a bowl, and eventually closes into a sphere. This is **positive Gaussian curvature**. It is

the geometry of balls, of amigurumi heads, of planets. Every crocheter who has made a sphere for a stuffed animal has constructed a surface of positive curvature.

The relationship is elegantly simple:

Increases per Round	Curvature	Surface	Crochet Example
Exactly right (6 for sc)	Zero	Flat plane	Dishcloth, flat coaster
Too many (>6 for sc)	Negative	Hyperbolic	Ruffled scarf, coral reef
Too few (<6 for sc)	Positive	Spherical	Amigurumi head, bowl

This connects to a deep result in mathematics: the **Gauss-Bonnet theorem**, which links the total curvature of a surface to its **Euler characteristic** (a topological invariant). The Euler characteristic is calculated as $V - E + F$, where V is the number of vertices, E is the number of edges, and F is the number of faces of the surface. For a sphere, the Euler characteristic is 2. For a flat torus (like a donut), it is 0. For a surface with a handle, it decreases.

When you crochet a closed sphere (an amigurumi ball), the total Gaussian curvature over the entire surface equals 4π — exactly as the Gauss-Bonnet theorem predicts for a surface with Euler characteristic 2. Your hands have proven a theorem in differential geometry, one stitch at a time.

2. The Stitch Network as a Graph

Now turn your crocheted fabric over and look at the back. Better yet, look at the front with your crafter's eye. Every stitch sits in a specific position, connected to the stitches below it (where the hook was inserted), beside it (the previous and next stitches in the row), and above it (the stitches that will be worked into it in the next row).

This is a **graph** in the mathematical sense — a collection of **nodes** (vertices) connected by **edges** (links). Each stitch is a node. Each connection between stitches is an edge. The crocheted fabric is not just *like* a network. It *is* a network, physically constructed from yarn.

Let us examine the properties of this graph:

Degree of a node. In graph theory, the **degree** of a node is the number of edges connected to it. In a standard row of single crochet, each stitch connects to: the stitch below (1 edge), the stitch to the left (1 edge), the stitch to the right (1 edge), and eventually the stitch above (1 edge). That gives a typical degree of 3 or 4, depending on position and whether the next row has been worked yet. Edge stitches have lower degree (fewer neighbors). Increase stitches create nodes where two stitches share one parent, changing the local degree distribution.

Regularity. A flat piece of single crochet worked back and forth creates a remarkably **regular graph** — almost like a grid, with slight offsets from the staggered row structure. This regularity is what gives stockinette-like fabrics their uniform appearance. When you introduce pattern stitches, you break this regularity in controlled ways.

Clustering. When you work stitches that involve inserting the hook into the same stitch multiple times (shell stitches, for example: 5 dc in one stitch), you create a **cluster** — a group of nodes that are all connected to the same parent node. This is a high-clustering-coefficient neighborhood in the graph, and it produces the fan-like shapes that shell stitch patterns are known for.

Comparing to neural network graphs. In an artificial neural network, nodes are neurons, edges are connections (synapses), and each connection has a weight. The network is typically organized in layers: input, hidden, and output. In a crocheted fabric, the rows are the layers. The foundation chain is the input layer. Each subsequent row is a hidden layer that transforms the information (the structure) from the previous layer. The final row — or the bind-off edge — is the output layer.

This is not just an analogy. The mathematical structure is genuinely similar. Both are directed graphs (in crochet, the direction flows from foundation to final row). Both have layered architecture. Both can have varying connectivity patterns (dense layers, sparse layers, skip connections). Both transform structure through local operations applied at each node.

3. Neural Network Topology and Fabric Topology

Let us push this parallel further, because it reveals something deep about how Active Inference connects topology and computation.

Feedforward architecture. A feedforward neural network passes information in one direction: input to output, layer by layer, no loops. Flat crochet worked in rows is feedforward. Row 1 feeds into Row 2, which feeds into Row 3. Information (structural pattern, stitch count, shaping) flows in one direction. You never go back and modify a completed row from a later one — at least, not without frogging.

Recurrent architecture. A recurrent neural network has loops — output from later layers feeds back into earlier layers. Crochet worked **in the round** is recurrent. When you join the last stitch of a round to the first stitch with a slip stitch, you create a loop in the graph. The end of the layer connects back to its beginning. Information circulates. This is why crocheting in the round produces a qualitatively different kind of fabric — seamless, continuous, with a different kind of structural integrity. It is also why hats, amigurumi, and mandalas have a fundamentally different topology than scarves and blankets.

Skip connections. In deep neural networks, **skip connections** (also called residual connections) allow information to jump over one or more layers. In crochet, working into stitches from two or more rows below — front post stitches, back post stitches, spike stitches — creates skip connections. The current row reaches past the immediately preceding row to connect with an earlier layer. Cable patterns and textured stitches frequently use this architecture.

The Markov blanket as a topological boundary. Here is where topology and Active Inference meet most directly. In Active Inference, the **Markov blanket** is the boundary that separates a system's internal states from external states. We discussed this in Course 1 as the working edge of the fabric. But now we can see it topologically.

The Markov blanket is a **topological boundary** — it divides the surface of the fabric into an interior region (completed work) and an exterior region (unworked yarn and environment). In topology, a boundary is a special kind of edge: the set of points where "inside" and "outside" meet. The working row in crochet is precisely this. It is a one-dimensional curve (the row of live stitches) that separates the two-dimensional surface of the fabric into distinct regions.

In a flat piece worked in rows, the Markov blanket is an open curve (the working row has two endpoints — the edges of the fabric). In a piece worked in the round, the Markov blanket is a **closed curve** — a loop with no endpoints. This topological difference has real consequences.

A closed Markov blanket (working in the round) creates a system with no edges to unravel from. An open Markov blanket (working in rows) has vulnerable endpoints.

The generative model in Active Inference is the system's internal model of how sensory observations are generated. For a crocheter, the generative model includes the pattern, the understanding of how stitches connect, and the expectation of what the fabric should look like. The topology of the fabric — its curvature, its connectivity, its boundary structure — is encoded in this generative model. When the crocheter expects a flat circle and sees ruffling, that is **prediction error**: the generative model predicted zero curvature but the fabric exhibits negative curvature. The crocheter's response — reducing increases on the next round — is **active inference**: adjusting actions to minimize the discrepancy between the model's predictions and the observed topology.

Applications

The intersection of crochet topology, neural network structure, and Active Inference has produced some remarkable projects:

- **The Crochet Coral Reef.** Margaret and Christine Wertheim's Institute For Figuring has organized a global project where thousands of crocheters create hyperbolic forms to build an enormous crocheted coral reef. Each ruffled, branching, curling piece is a surface of negative curvature, and the project as a whole visualizes both marine biology and hyperbolic geometry. The reef demonstrates that crochet can model organic forms that arise from the same topological principles that govern biological growth — organisms whose cells divide at rates that produce negative curvature, just as a crocheter's increases produce hyperbolic ruffles.
- **The Crocheted Lorenz Manifold.** Mathematicians Hinke Osinga and Bernd Krauskopf crocheted a physical model of the Lorenz manifold — the surface traced out by the famous butterfly-shaped Lorenz attractor from chaos theory. The pattern required precise control of increases and decreases to create a surface with the exact curvature profile of the mathematical object. The crocheted model, made from 25,511 stitches, allowed researchers to hold and examine a geometric object that had previously existed only as a computer rendering.

- **Neural Network Visualization.** Educators and artists have begun using crocheted networks to teach neural network architecture. Each stitch node can be a different color to represent activation levels. Rows of stitches become layers of the network. Increase stitches (where one node fans into many) represent diverging connections, while decrease stitches (where many nodes merge into one) represent converging connections. The physical, three-dimensional nature of the crocheted network makes the architecture tangible in a way that diagrams on a whiteboard cannot.
- **Crocheted Klein Bottles and Mobius Strips.** Topology enthusiasts have crocheted Klein bottles (surfaces with no inside or outside) and Mobius strips (surfaces with one continuous side). These objects, which are difficult to visualize from equations alone, become intuitive when you hold them in your hands. The crochet construction process — building the surface row by row — makes the topological twist visible and graspable.

Conclusion

We began with a simple observation: crochet builds surfaces. But surfaces have topology, and topology connects to computation in deep ways. The flat disc, the hyperbolic ruffle, and the spherical bowl are not just craft objects — they are embodiments of Gaussian curvature, built by hands that understand geometry at a level that precedes formal mathematics. The stitch network of a crocheted fabric is a graph with the same structural features as a neural network: layers, connectivity, clustering, and directed information flow. And the Markov blanket of Active Inference — the boundary between system and environment — turns out to be a topological boundary, a curve that divides the surface of the fabric into inside and outside.

Crochet is a universal language for these ideas. It lets you hold topology in your hands, trace a neural network with your finger, and feel a Markov blanket dissolve when a loop slips off the hook. In the next module, we turn from the fabric to the crocheter and ask: what kind of agent designs and builds these networks?

Key Terms

Term	Crochet Meaning	Active Inference / Topology Meaning
Surface	The fabric created by interconnected stitches	A two-dimensional manifold embedded in three-dimensional space
Gaussian curvature	The shape of fabric determined by increase/decrease rate	The intrinsic curvature at a point on a surface (product of principal curvatures)
Euler characteristic	The global shape type of a finished object (sphere, torus, flat)	Topological invariant $V - E + F$; classifies surfaces
Node	A single stitch in the fabric	A vertex in a graph; a neuron in a neural network
Edge	The yarn connection between stitches	A link in a graph; a synapse in a neural network
Degree	How many connections a single stitch has to its neighbors	Number of edges incident to a node
Feedforward	Working in rows, one direction, no loops	Neural network architecture with no recurrent connections
Recurrent	Working in the round, where end of round connects to beginning	Neural network architecture with feedback loops
Skip connection	Spike stitch or post stitch reaching past the previous row	Residual connection that bypasses one or more layers
Markov blanket	The working edge — live loops, hook, and working row	The topological boundary separating internal from external states

Term	Crochet Meaning	Active Inference / Topology Meaning
Generative model	The pattern and the crocheter's understanding of stitch structure	The internal model of how observations are generated
Prediction error	Fabric not matching expectations (ruffling when flat was intended)	Discrepancy between predicted and observed sensory states
Hyperbolic surface	Ruffled fabric from too many increases	Surface of constant negative Gaussian curvature